Antiderivatives and Indefinite Integrals

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Basic antiderivatives

Find each indefinite integral:

a $\int (3x^2 - 4x + 5)\,dx$

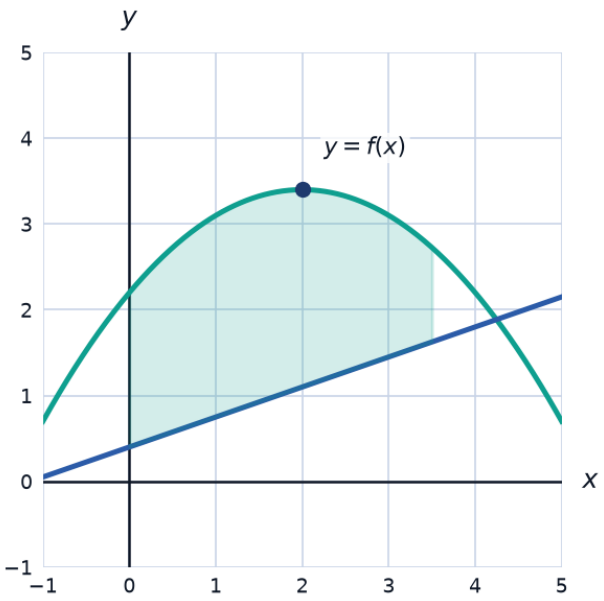
b $\int \frac{1}{x^2}\,dx$

c $\int \sqrt{x}\,dx$

d $\int \sec^2 x\,dx$

2 [Foundational] • Target: Basic antiderivatives

Find $\int (4\sin x + e^x)\,dx$.



3 [Foundational] • Target: Using initial conditions

Find $f(x)$ given that $f'(x) = 6x^2 - 2$ and $f(1) = 5$.

4 [Foundational] • Target: Using initial conditions

A particle moves along a line with velocity $v(t) = 3t^2 - 6t$ m/s. If its position at $t = 0$ is $s(0) = 4$ m, find its position function $s(t)$.

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5 [Foundational] • Target: Using initial conditions

An object is dropped from rest, so its acceleration is $a(t) = -9.8$ m/s^2.

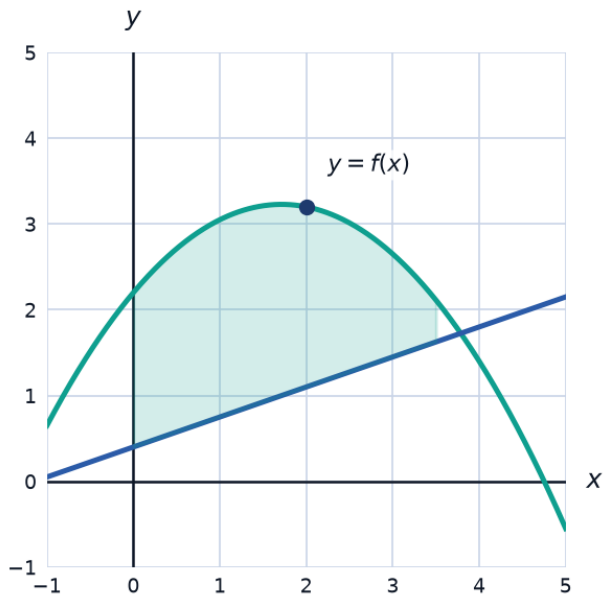
a Find the velocity function $v(t)$.

b If the object starts at height $s(0) = 45$ m, find its position function $s(t)$.

c Find how long it takes to hit the ground ($s = 0$), to two decimal places.

6 [Foundational] • Target: More antiderivatives

Find the general antiderivative of $f(x) = (x + 1)^3$ (expand first, or use the power rule on the whole expression, and state which method you used).



7 [Foundational] • Target: Motion applications

A rocket's acceleration is $a(t) = 6t$ m/s^2, with $v(0) = 0$ and $s(0) = 0$.

a Find $v(t)$.

b Find $s(t)$.

c Find the distance traveled in the first 4 seconds.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Antiderivatives of composite expressions

Find $\int (2x - 1)^5 \, dx$, treating $(2x - 1)$ as a linear inner function.

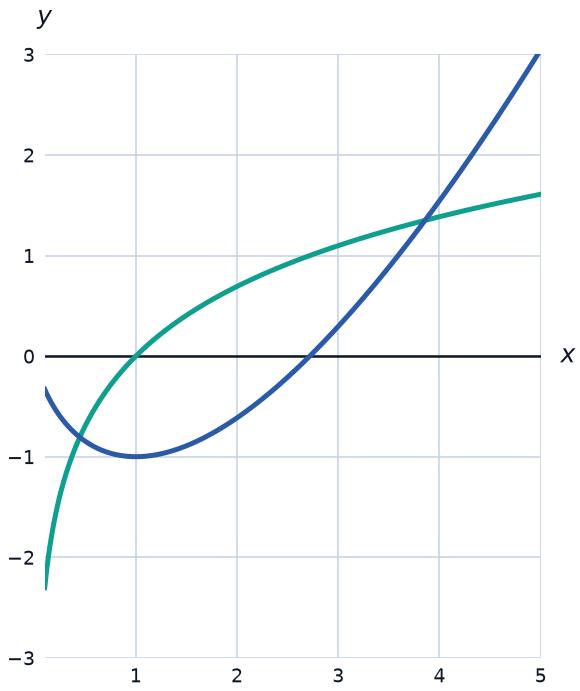
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9 [Intermediate] • Target: Antiderivatives of composite expressions

Find $\int \sin(3x)\,dx$ and $\int e^{5x}\,dx$.

10 [Intermediate] • Target: Verifying antiderivatives by differentiation

Verify that $F(x) = x\ln x - x$ is an antiderivative of $f(x) = \ln x$ by differentiating $F(x)$.



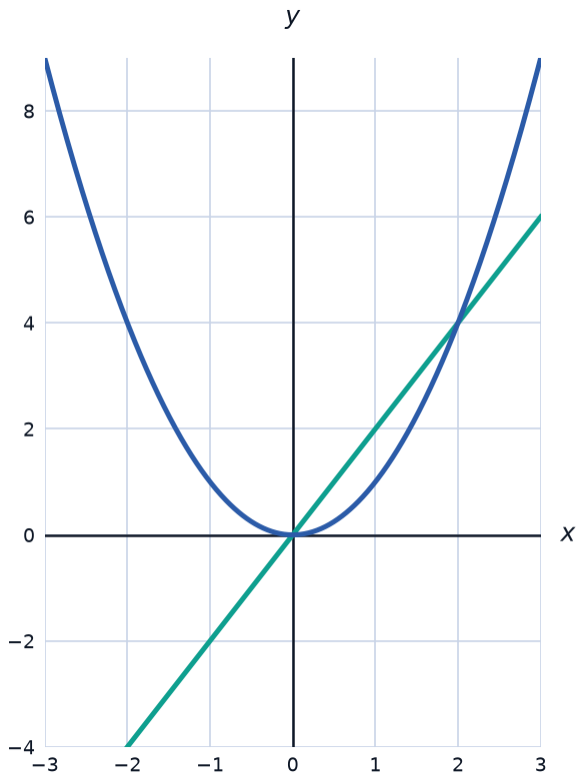
11 [Intermediate] • Target: A projectile motion application

A ball is thrown upward with initial velocity 15 m/s from a height of 2 m, so its acceleration is $a(t) = -9.8$ m/s^2. Find its velocity function $v(t)$ and position function $s(t)$.

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12 [Intermediate] • Target: Visualizing an antiderivative family

The graph shows $f(x) = 2x$ and one member of its antiderivative family, $F(x) = x^2$. Every other antiderivative of f is a vertical shift of F.



a Write the general antiderivative of $f(x) = 2x$.

b If a specific antiderivative passes through $(1, 4)$, find its constant C.

13 [Intermediate] • Target: Antiderivatives requiring algebraic rewriting first

Find $\int \frac{x^3 + 2x}{x}\,dx$ by simplifying the integrand before integrating.

14 [Intermediate] • Target: Antiderivatives requiring algebraic rewriting first

Find $\int (x + 1)(x - 3)\,dx$ by expanding the product first.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

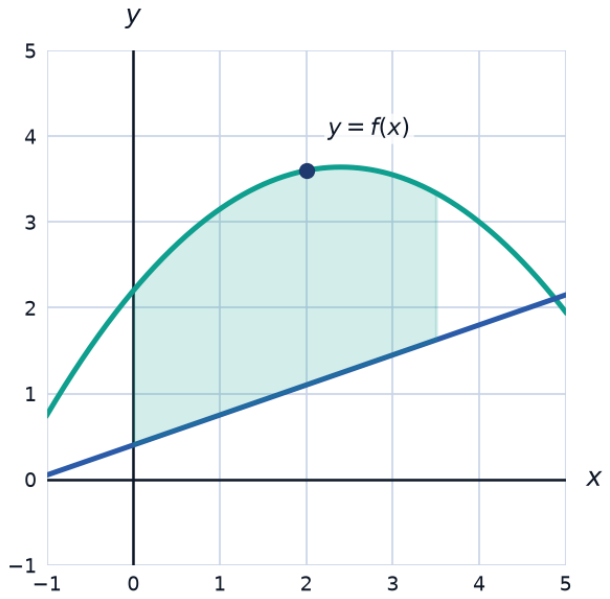
15 [Challenging] • Target: Basic antiderivatives • Cross-Lesson: 12.02 Power Series and Taylor Series

Consider a particle whose velocity vector $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle$ satisfies the differential equation $y'' - 2y' + y = 0$ and the initial conditions $x(0) = 1, y(0) = 2, x'(0) = 0, y'(0) = 1$. Furthermore, the particle's path is such that the magnitude of its acceleration vector is always equal to the magnitude of its velocity vector, i.e., $||\mathbf{a}(t)|| = ||\mathbf{v}(t)||$. Find the value of $x(1)$.

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16 [Challenging] • Target: Using initial conditions • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Consider the region R bounded by the curve $y = \frac{1}{x^2+1}$, the x-axis, and the vertical lines $x = 0$ and $x = 1$. A solid is formed by rotating R about the y-axis. Determine the volume of this solid using the method of cylindrical shells. Express your answer in terms of π.



17 [Challenging] • Target: More antiderivatives • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Composite series: $e^{\sin x}$ up to x^3. Using $\sin x = x - x^3/6 + \cdots$ inside $e^u = 1 + u + u^2/2 + \cdots$:.

18 [Challenging] • Target: Motion applications • Cross-Lesson: 15.01 Double Integrals

Volume of cone: radius r, height h, using disc method:.

19 [Challenging] • Target: Antiderivatives of composite expressions • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Volume of region $y \geq x^2$, $y \leq 4$ rotated about $y = 4$ using washers:.

20 [Challenging] • Target: Verifying antiderivatives by differentiation • Cross-Lesson: 2.01 Evaluating Limits

Error bound of Simpson's rule on $[a, b]$ with n intervals:.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The Power Rule for Integration

Evaluate $\int (6x^3 - 4x + 5)\,dx$.

a. $\frac{3}{2}x^4 - 2x^2 + 5x + C$

b. $2x^4 - 2x^2 + 5x + C$

c. $18x^2 - 4 + C$

d. $\frac{3}{2}x^4 - 4x^2 + 5x + C$

22 [Foundational] • Target: The Power Rule for Integration

Find $\int \left(x^{2/3} + \frac{1}{x^3}\right) dx$.

a. $\frac{3}{5}x^{5/3} - \frac{1}{2x^2} + C$

b. $\frac{5}{3}x^{5/3} - \frac{1}{4x^4} + C$

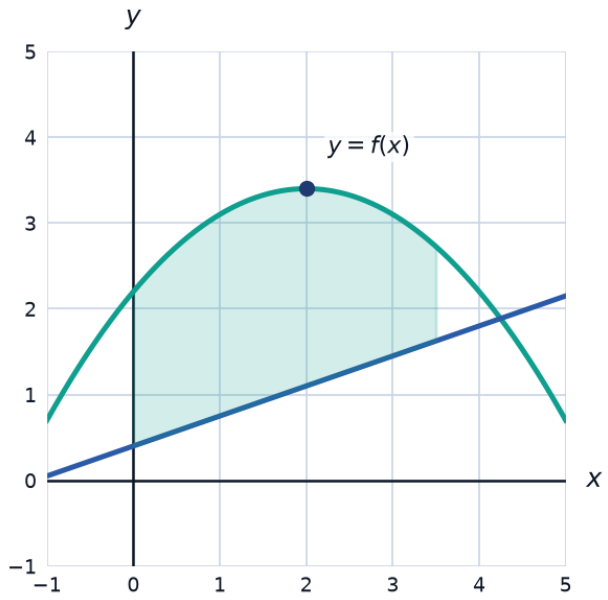
c. $\frac{3}{5}x^{5/3} + \frac{1}{2x^2} + C$

d. $\frac{2}{5}x^{5/3} - \frac{2}{x^2} + C$

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23 [Foundational] • Target: The Power Rule for Integration

Evaluate $\int \frac{3x^4 - 2x}{x^2}\,dx$.



a. $\frac{3}{3}x^3 - 2\ln|x| + C = x^3 - 2\ln|x| + C$

b. $x^3 - \frac{2}{x} + C$

c. $3x^3 - 2x + C$

d. $x^3 - 2x^{-1} + C$

24 [Foundational] • Target: Trigonometric and Exponential Antiderivatives

Evaluate $\int (3\sin x + 4\cos x)\,dx$.

a. $-3\cos x + 4\sin x + C$

b. $3\cos x - 4\sin x + C$

c. $3\cos x + 4\sin x + C$

d. $-3\cos x - 4\sin x + C$

25 [Foundational] • Target: Trigonometric and Exponential Antiderivatives

Evaluate $\int (5e^x + \sec^2 x)\,dx$.

a. $5e^x + \tan x + C$

b. $5e^x - \tan x + C$

c. $5e^x + \sec x \tan x + C$

d. $e^x + \tan x + C$

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26 [Foundational] • Target: Trigonometric and Exponential Antiderivatives

Find $\int (2^x + \csc x \cot x)\, dx$.

a. $\frac{2^x}{\ln 2} - \csc x + C$

b. $2^x \ln 2 - \csc x + C$

c. $\frac{2^x}{\ln 2} + \csc x + C$

d. $2^x - \cot x + C$

27 [Foundational] • Target: Initial Value Problems

Find $f(x)$ given $f'(x) = 4x - 3$ and $f(1) = 6$.

a. $f(x) = 2x^2 - 3x + 7$

b. $f(x) = 2x^2 - 3x + 5$

c. $f(x) = 4x^2 - 3x + 5$

d. $f(x) = 2x^2 - 3x$

28 [Foundational] • Target: Initial Value Problems

A curve satisfies $\frac{dy}{dx} = 3x^2 + 2x$ and passes through $(2, 10)$. Find its equation.

a. $y = x^3 + x^2 - 2$

b. $y = x^3 + x^2 + 2$

c. $y = 3x^3 + 2x^2 - 22$

d. $y = x^3 + x^2$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Initial Value Problems

Given $f''(x) = 6x$, $f'(0) = 4$, and $f(0) = 1$, find $f(x)$.

a. $f(x) = x^3 + 4x + 1$

b. $f(x) = 3x^3 + 4x + 1$

c. $f(x) = x^3 + 2x + 1$

d. $f(x) = x^3 + 4$

30 [Intermediate] • Target: Rectilinear Motion and Acceleration

A particle moves with acceleration $a(t) = 6t - 4$. If its initial velocity $v(0) = 5$, find $v(t)$.

a. $v(t) = 3t^2 - 4t + 5$

b. $v(t) = 6t^2 - 4t + 5$

c. $v(t) = 3t^2 - 4t$

d. $v(t) = 3t^2 + 4t + 5$

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31 [Intermediate] • Target: Rectilinear Motion and Acceleration

If a ball is thrown upward with $v(0) = 48$ ft/s and $a(t) = -32$ ft/s^2 from ground level $s(0) = 0$, find its maximum height.

a. 36 ft

b. 48 ft

c. 64 ft

d. 32 ft

32 [Intermediate] • Target: Rectilinear Motion and Acceleration

A car decelerates at a constant $a = -4$ m/s^2 from $v_0 = 24$ m/s. How far does it travel before coming to a stop?

a. 72 m

b. 48 m

c. 96 m

d. 36 m

33 [Intermediate] • Target: Initial Value Problems

Area of triangle with vertices at origin, $(3, 0)$, $(0, 4)$ using integration:

a. 3

b. 12

c. 6

d. π

34 [Intermediate] • Target: Rectilinear Motion and Acceleration

Volume of revolution using shells: $y = \cos(x^2)$, $x \in [0, \sqrt{\pi/2}]$, rotated about y-axis:

a. 2π

b. $\pi/2$

c. π

d. $\sqrt{\pi}$

35 [Intermediate] • Target: Basic antiderivatives

Approximate $\int_0^1 \frac{\sin x}{x} dx$ using power series (first three terms):

a. $1 - 1/3 + 1/10$

b. $1 - 1/18 + 1/600$

c. $1 - 1/6 + 1/20$

d. $1 - 1/18$

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36 [Intermediate] • Target: Using initial conditions

Surface area of sphere radius R using surface area of revolution:

a. $4\pi R^2$

b. $3\pi R^2$

c. $2\pi R^2$

d. πR^3

37 [Intermediate] • Target: More antiderivatives

Power series integration: $\int_0^x \frac{1}{1+t^2}dt = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} =$

a. $\arctan x$

b. $\ln(1+x)$

c. $\sin x$

d. $x/(1+x^2)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Motion applications • Cross-Lesson: 15.01 Double Integrals

Numerical integration: Simpson's rule uses polynomials of degree:

a. 1

b. 3

c. 2

d. 4

39 [Challenging] • Target: Antiderivatives of composite expressions • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Volume of a sphere radius R using disc method:

a. $\frac{4}{3}\pi R^3$

b. $\frac{4}{3}\pi R^2$

c. $4\pi R^2$

d. $2\pi R^3$

40 [Challenging] • Target: Verifying antiderivatives by differentiation • Cross-Lesson: 2.01 Evaluating Limits

Volume integral: using spherical coordinates, $dV =$

a. $dr\,d\theta\,d\phi$

b. $r^2 \sin\phi\,dr\,d\theta\,d\phi$

c. $r\sin\theta\,dr\,d\theta\,d\phi$

d. $r^2\,dr\,d\theta\,d\phi$

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41 [Challenging] • Target: Antiderivatives of reciprocal and rational-power functions • Cross-Lesson: 2.02 Continuity

LIATE rule for IBP: first choice for u among x^2, e^x, $\ln x$:

a. e^x

b. x^2

c. $\ln x$

d. any of them

42 [Challenging] • Target: A projectile motion application • Cross-Lesson: 3.01 The Derivative from First Principles

The Padé approximant $[m/n]$ for $f(x)$ is a rational function $P_m(x)/Q_n(x)$ matching the Taylor series to order $m+n$. It typically converges more:

a. slowly than Taylor

b. only at 0

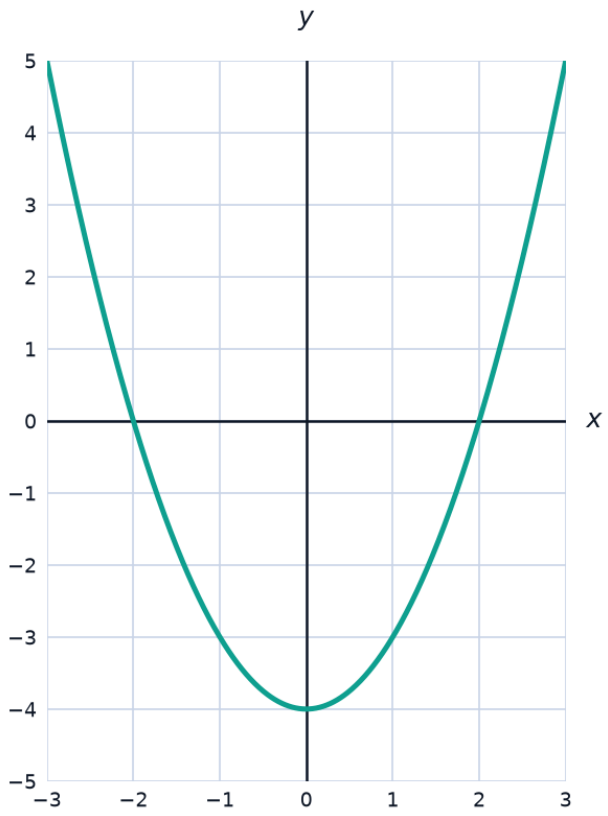
c. rapidly than Taylor for poles

d. never

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43 **[Challenging]** • **Target: Visualizing an antiderivative family** • **Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain**

The region bounded by the curve $y = x^2 - 4$ and the x-axis is rotated about the x-axis. If the volume of the resulting solid is $V = \pi\int_a^b (x^2 - 4)^2 dx$, find the value of $b - a$.



a. 2

b. 4

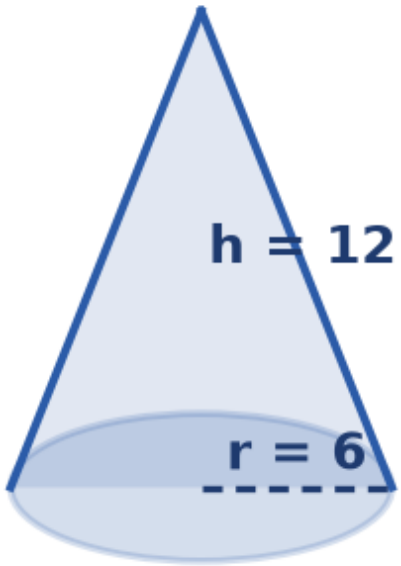
c. 8

d. 16

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44 [Challenging] • Target: Antiderivatives requiring algebraic rewriting first • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Water drains from a cone-shaped tank (radius 6, height 12) at $\frac{dV}{dt} = 1$ m^3/min. Using $V = \frac{1}{3}\pi r^2 h$ with $r = \frac{6}{12}h$, find $\frac{dh}{dt}$ when $h = 5$.



a. $-\frac{4}{25\pi}$

b. $\frac{8}{25\pi}$

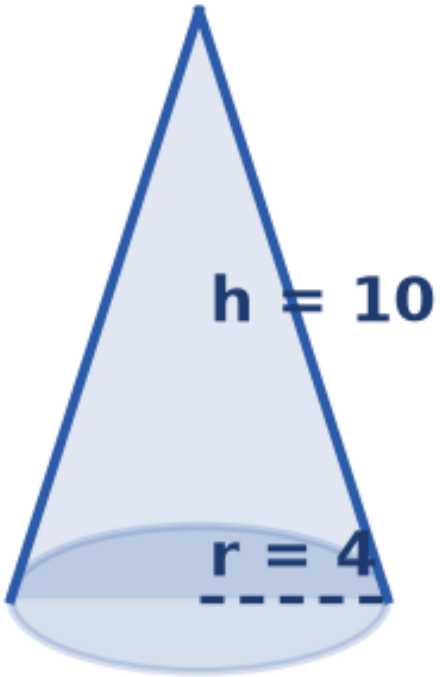
c. $\frac{2}{25\pi}$

d. $\frac{4}{25\pi}$

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45 [Challenging] • Target: The Power Rule for Integration • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Water drains from a cone-shaped tank (radius 4, height 10) at $\frac{dV}{dt} = 2$ m^3/min. Using $V = \frac{1}{3}\pi r^2 h$ with $r = \frac{4}{10}h$, find $\frac{dh}{dt}$ when $h = 2$.



a. $\frac{25}{4\pi}$

b. $-\frac{25}{8\pi}$

c. $\frac{25}{16\pi}$

d. $\frac{25}{8\pi}$

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